TREE-BASED MULTI-OUTPUT REGRESSION FOR NEAR REAL-TIME PERFORMANCE PREDICTION IN VR FLIGHT TRAINING

YIJUN WU

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE
SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
TILBURG UNIVERSITY

STUDENT NUMBER

2066623

COMMITTEE

dr. Phillip Brown
dr. Silvia-Laura Pintea

LOCATION

Tilburg University
School of Humanities and Digital Sciences
Department of Cognitive Science &
Artificial Intelligence
Tilburg, The Netherlands

DATE

May 21, 2026

WORD COUNT

7000

ACKNOWLEDGMENTS

Some room for acknowledgements.

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Abstract

This study investigates the predictive accuracy of task performance during virtual reality landing tasks in near real time from multimodal physiological and simulator-derived features using tree-based machine learning models. The objective is to assess whether such models can support automated, continuous monitoring in pilot training scenarios. Analyses are based on the publicly available CogPilot dataset, which contains synchronized physiological signals and aircraft behavior data from participants following an instrument landing system procedure in an immersive VR flight simulator. ECG and respiration signals were preprocessed to derive heart-rate and breathing features within sliding 10-second windows and combined with aircraft control variables to form second-by-second observations. Random Forest, XGBoost, and CatBoost multi-output regressors were trained to predict performance errors and were evaluated under subject-level and temporal generalization settings, with and without one- and two-step lagged performance features. Models with lagged features achieved high predictive accuracy compared with models using only non-lagged features. The best CatBoost configurations attained mean coefficients of determination close to 0.98 across settings and preserved the correlation structure between component and total errors. Feature-importance analyses showed that very recent aircraft control and performance history dominated predictions, while physiological features provided modest additional value.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source: The data used in this thesis come from the publicly available CogPilot dataset, published on PhysioNet [1] by [2], who remain the owners of the data. No new data were collected for this project, and the research did not involve direct interaction with human participants or animals. The dataset is anonymized and was obtained through an online request and download from the PhysioNet repository in accordance with its terms of use.

Code: All preprocessing, modeling, and analysis code was written by the author in Python, using the libraries NeuroKit2, pandas, NumPy, scikit-learn, XGBoost, CatBoost, Matplotlib, and seaborn. The thesis code can be accessed through the GitHub repository following the link https://github.com/Yijun-cat/Bachelor_Thesis_Project.git. Where code patterns followed standard examples from package documentation or tutorials, these were adapted and integrated into the author's own pipelines; no proprietary or unpublished code from other studies was directly reused.

Figures: All figures in this thesis were generated by the author using the analysis code developed for this project, based on the dataset and derived model outputs. No externally created images or figures were reused.

Technology: During the preparation of this work, the author used Perplexity [3], powered by GPT-5.1 [4], in order to assist with brainstorming, language polishing, and checking the clarity and consistency of the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the final text of the thesis. Standard spelling and grammar checking tools integrated in the word processing and LaTeX environment were also used. No additional tools or services were used for typesetting beyond the official Tilburg University LaTeX template and its associated reference management workflow.

2 INTRODUCTION

Training professionals in high-stake environments requires the acquisition of complex psychomotor and cognitive skills and attaining expert-level competency proves challenging due to strict safety concerns and constrained training resources [5]. In the aviation industry, pilot training plays a vital role and is directly linked to both flight safety and the effective execution of flight operations [6]. However, the current training framework is largely instructor-led and can lead to high instructor workload, subjective trainee evaluation, and a lack of real-time feedback on risky behaviors [6]. Multiple studies have examined the integration of machine learning (ML) techniques with virtual reality (VR) to address these challenges; nevertheless, the adoption of ML in pilot training and education remains comparatively limited [7]. ML approaches that leverage physiological signals from trainee pilots can be integrated into legacy training equipment and simulators [7], and they enable more precise and continuous error measurement, reducing the need for labor-intensive evaluations by human instructors [8]. Building on these developments, the present project investigates near real-time task performance prediction using ML models trained on multimodal physiological and simulator-derived data, with the goal of informing the design of adaptive and scalable pilot training systems.

From a scientific perspective, this thesis advances research on physiology-based flight performance modeling by systematically assessing tree-based regression methods for the continuous prediction of multiple performance-error metrics in a VR landing task. Psychophysiological signals have been used in prior work to monitor pilot performance in real time and to forecast fatigue [9, 10, 11], demonstrating the feasibility of physiology-informed performance models in aviation contexts. Many of these studies, however, have focused on either physical cockpits with 2D simulator displays or real-flight scenarios, and performance monitoring and modeling within fully immersive VR training environments remain comparatively underexplored. Existing work in VR-integrated simulators has often relied on single-modal physiological inputs when applying machine learning to analysis of pilot performance [12, 13]. Given these limitations, the present thesis investigates multimodal physiological and simulator-derived features for performance prediction in an integrated VR environment, thereby providing empirical evidence on how such models can support near real-time, adaptive training feedback.

Flight simulators are becoming increasingly important for training and research because of growing concerns about flight safety, shortages of qualified pilots and instructors, and rapid technological developments in the aviation industry [14]. Virtual reality has further advanced simulator-based pilot training by integrating VR headsets into existing platforms and providing high-fidelity three-dimensional flight scenarios, which offer immersive and realistic virtual cockpits while improving cost-effectiveness, operational safety, and overall training efficiency [14]. Despite these advantages, current simulator-based training still depends extensively on human instructors and evaluators for continuous performance monitoring and adaptive difficulty adjustment, which constrains training output when instructor resources are limited [7]. Adaptive automated training systems that continuously monitor pilot performance and dynamically adjust task difficulty could help alleviate these constraints and reduce the operational burden on active pilots who serve as instructors [8]. From a societal perspective, by enabling more accurate and timely prediction of task performance from multimodal physiological and simulator-derived data, the models examined in this thesis can inform the design of such systems and support safer and more scalable pilot training.

The central research question of this thesis is therefore:

How accurately can task performance during virtual reality pilot training tasks be predicted in near real time using tree-based machine learning models trained on multimodal physiological and simulator-derived features?

In this thesis, the research question is addressed by training Random Forest, XGBoost, and CatBoost regressors to predict second-by-second glide slope, localizer, airspeed, and total error during VR landing tasks from electrocardiography- (ECG-) and respiration-derived features together with simulator variables, using both subject-level and temporal cross-validation. In both evaluation settings, models using only physiological and aircraft features achieve low predictive accuracy, whereas adding one- and two-second lagged flight performance errors yields a large improvement in all evaluation metrics. Feature-importance analyses indicate that very recent performance history is the dominant predictor of near-future errors, while physiological features provide additional but comparatively modest contributions, suggesting that tree-based models with simple lagged inputs can support highly accurate, near real-time monitoring of task performance in VR-based pilot training.

3 RELATED WORK

This section reviews prior work relevant to VR pilot training, physiological measurement, and machine learning-based performance modeling. The aim is to position the present study within the broader literature and to highlight how existing approaches inform, but also differ from the goals of this thesis.

3.1 *VR in Aviation Training*

Virtual reality (VR) has increasingly been adopted across multiple domains of aviation training, ranging from ab initio pilot instruction to aircraft maintenance and service roles [15, 16, 17]. Studies with student pilots have evaluated VR flight simulators as primary or supplementary training devices and typically compare them to desktop-based or traditional aviation training devices [16, 18]. These developments highlight its potential as a general-purpose medium for high-fidelity simulation in aviation.

The effectiveness of VR in pilot training has been assessed in a number of empirical studies. For ab initio pilot training, Hight *et al.* [16] conducted a quasi-experimental pilot study with civilian pilots, comparing desktop 2D simulations and head-mounted VR simulations while keeping other training factors identical. They found no significant differences in academic performance between the VR and 2D conditions, based on module and final scores, and also collected subjective ratings of the learning materials. Their findings suggest that VR does not hinder skill acquisition, and that students in both conditions rated simulation-based training as highly beneficial and preferred it over other learning media [16]. In another study, Guthridge and Clinton-Lisell [18] evaluated beginning-level instrument pilots' pretest and post-test performance on a visual traffic pattern task after trained in either VR simulator, a PC-based simulator, or no simulator. The results showed that both the VR and PC-based training groups improved significantly on flight maneuvers relative to the group without simulator training, with the VR group performing broadly comparably to the PC-based group [18]. Students also perceived VR as an acceptable and immersive training technology [18].

These studies suggest that immersive VR simulators can support training outcomes comparable to those achieved with 2D desktop or PC-based simulators. Students generally perceive both 2D and VR simulations as beneficial and often prefer simulator-based practice over more traditional learning materials [16, 18], even though current VR systems still present technical, implementation, and integration challenges [17]. However, existing evaluation methods mostly rely on aggregated academic performance or discrete measures of maneuver accuracy rather than continuous, time-resolved performance metrics or physiology-informed models. This methodological emphasis underscores a persisting gap in the literature and motivates the present thesis, which seeks to integrate VR flight tasks with multimodal physiological monitoring and machine-learning-based performance prediction.

3.2 *Physiological Measures of Workload in Pilot Training*

A number of studies have investigated how ECG- and respiration-based measures reflect cognitive load and stress in simulated flight tasks. Korner *et al.* [19] examined heart rate as an index of mental workload and performance in VR landing tasks, and found that mean heart rate increased with task difficulty, decreased across repeated runs of the same difficulty, and rose near touchdown compared with the start of the flight, while correlating positively with subjective workload and negatively with performance. These results indicate that heart rate can support real-time task-difficulty adaptation to keep trainees within an optimal workload band [19]. Bruna *et al.* [20] investigated ECG and respiration during simulated emergency landings and related them to pilots' subjective stress ratings, and reported that ECG-derived heart-rate variability parameters showed no significant changes across flight phases, whereas several respiration features, including average breathing frequency and low-frequency respiratory power, increased during approach and landing [20]. These findings suggest that breathing rate and spectral power are sensitive markers of elevated workload and stress in high-demand piloting scenarios [20].

While these studies demonstrate that physiological data provide useful indicators of mental workload and stress, they primarily focus on analyses of physiology averaged over discrete flight phases or conditions and treat physiology as an outcome or indicator of task demand rather than as input to models predicting continuous performance. In contrast, the present thesis incorporates ECG- and respiration-derived features, together with simulator-derived variables, into tree-based regression models that aim to predict performance errors in near real time. By evaluating how well these models generalize across subjects and over future time segments within runs, the thesis investigates the extent to which multimodal physiological signals can support continuous, sample-wise prediction of pilot performance.

3.3 *Machine Learning for VR Flight Performance Prediction*

Several studies have explored machine learning models trained on different physiological and simulator-derived signals to support the development of adaptive VR-integrated flight simulators for pilot training. A commonly used dataset in this context is the CogPilot [2] corpus, which contains multimodal physiological and simulator data from participants performing VR-based landing tasks. Caballero *et al.* [7] used this dataset to classify flight difficulty from multimodal physiological signals, training an AdaBoost-based pipeline on features derived via functional principal component analysis and feature selection, and achieving around 87% accuracy with an Area Under the Curve (AUC) of 0.88. Building on the same dataset, Moore *et al.* [12] focused on eye-tracking and electrodermal activity signals for the same classification problem, combining the Minimally Random Convolutional Kernel Transform with a deep neural network; their model achieved an AUC of 0.91 while reducing the original time-series representation by approximately 99.7%. More recently, Barry *et al.* [8] proposed a multi-modal functional learning framework for landing-error regression on the same dataset, transforming high-frequency physiological and aircraft signals into tabular features using summary statistics and functional principal component analysis, applying feature selection with BorutaSHAP, and training several tree-based models to predict approach-error metrics.

Together, these studies demonstrate that multimodal time-series from VR flight tasks can be effectively leveraged by machine learning models for both flight-difficulty classification and offline prediction of landing performance, and they provide an important methodological foundation for intelligent instructor and evaluator systems. However, existing work predominantly targets post-hoc analysis using either discrete difficulty labels or aggregated performance scores, and does not address continuous, near real-time prediction of approach-error trajectories or systematically examine subject-level and temporal generalization behavior of different tree-based model families. The present thesis extends this line of research by applying and comparing tree-based multi-output regression models to predict performance metrics in near real time from multimodal physiological and simulator-derived features, and by explicitly analyzing their generalization across unseen pilots and across future time segments within runs.

4 METHODS

The following sections describe the dataset, the extraction of physiological and aircraft features in sliding time windows, and the construction of multi-output targets. They also provide a brief summary of the machine learning model training and the metrics used for evaluation.

4.1 *Dataset*

The CogPilot [2] dataset was used in this project and is publicly available via PhysioNet [1]. The dataset contains multimodal physiological, task performance, and interaction data collected from 35 participants who each performed flight–landing tasks at various difficulty levels. In a virtual reality environment, each participant followed an Instrument Landing System (ILS) approach and primarily relied on cockpit instrument readings to land an aircraft along a prescribed ideal flight path [2]. Each experimental session consisted of three blocks with four missions per block, resulting in twelve missions per participant, with one difficulty level per mission [2].

Multiple wearable sensors were used to record seven physiological modalities, with nominal sampling rates ranging from 128 to 1024 Hz [2]. These modalities comprise electromyography (EMG), accelerometry (ACC), electrodermal activity (EDA), electrocardiography (ECG), respiration (RESP), eye tracking (ETK), and photoplethysmography (PPG). A detailed overview of the functional data derived from each physiological signal is provided in Table 9 in Appendix A (page 24). Mission difficulty was manipulated via four distinct weather conditions [2]. According to the dataset description, flight performance measures were derived from the simulated aircraft states and grouped into three components [2]. The first two are the localizer and glide slope errors, defined as the angular deviations from the ideal horizontal and vertical landing trajectories [2], respectively, each ranging from -3 to 3 degrees. The third component is the absolute deviation from the ideal airspeed, where the raw airspeed error was divided by 10 so that the maximum error is mapped to the same -3 to 3 scale [2].

4.2 *Feature Engineering*

Recordings of ECG and RESP were selected as the physiological modalities for feature preprocessing and extraction. Heart rate (HR) derived from ECG can be used as a practical signal for real-time adaptation of task difficulty to maintain trainees within an optimal workload range [19], and respiratory behavior is a robust marker of cognitive processing, with mentally demanding tasks typically associated with an increased breathing rate (BR) and higher-frequency ventilation [21].

The raw ECG signal was first band-pass filtered between 8 and 20 Hz using a Butterworth filter, as this passband has been shown to be optimal for QRS complex detection with a favorable signal-to-noise ratio [22]. R-peaks were then detected in the cleaned signal using the QRS detection algorithm proposed by [22]. Subsequently, the signal-quality index (SQI) mechanism from [23] was applied to identify low-quality ECG segments, and HR was computed from the detected R-peaks only for 25 subjects with ECG recordings classified as having excellent quality.

For the RESP signals, an optimized breath detection algorithm developed by [24] was used to preprocess the respiratory traces and to derive breathing rate and respiratory peaks. This method has been shown to outperform conventional algorithms on breath detection in terms of receiver–operating–characteristic performance [24]. Respiratory amplitude was computed as the difference between each respiratory peak and its preceding trough.

Because ECG and RESP were recorded using the same sensing device, no additional timestamp alignment was required between these two signals. For each subject, difficulty level, and run number, physiological and aircraft features were extracted within a sliding 10-second window that was aligned to the performance timestamps. First, the performance, ECG, RESP, and aircraft data streams were converted from MATLAB time format to seconds and aligned by correcting for the offset between the start times of the performance and physiological recordings, yielding a common time axis for all signals. A 10-second window $(t - 10, t]$ was then defined and advanced in 1-second steps along the aligned performance timeline. For each window, the last available performance sample was used to assign the glide slope, localizer, airspeed, and total error as target variables at time t.

Within the same 10-second interval, the corresponding segments of the cleaned HR and respiration signals were selected, and the following time-domain features were computed: the mean, standard deviation, minimum, maximum, and range of HR; the mean and standard deviation of BR; and the mean and standard deviation of respiratory amplitude. In parallel, the mean aircraft elevation trim, aileron trim, and landing-gear position were computed over the window. Each window thus yielded a single feature vector, comprising physiological and aircraft features, paired with the contemporaneous performance errors, resulting in a second-by-second sequence of 10-second summary observations for subsequent modeling.

4.3 *Machine Learning Setup*

All predictive analyses were conducted on subject-level dataframes constructed from the 10-second windows described above. For each subject, difficulty level, and run number, physiological, aircraft, and performance information were aggregated into a feature matrix and a multi-output target matrix, where each row represents one time window and the targets correspond to glide slope, localizer, airspeed, and total error. Two feature configurations were considered. In the without-lagged-features condition, the feature set contained only physiological and aircraft features derived from the current window. In the with-lagged-features condition, the feature set was augmented with lagged performance measures corresponding to the four target variables at the preceding one and two seconds $(t-1, t-2)$, thereby enabling the models to exploit very recent performance history.

To quantify generalization, two cross-validation strategies were implemented that respect the hierarchical and temporal structure of the data. For subject-level generalization, leave-one-subject-out cross-validation was applied using the subject identifier as grouping variable, such that all windows from one participant were held out as test data in each fold and the models were trained on the remaining participants. For within-run temporal generalization, the final 60 seconds of data in each run of the subjects were excluded from the training–validation set, flagged as temporal test segments, and retained exclusively for final evaluation. On the remaining data , grouped 5-fold cross-validation was used, with groups defined as unique subject–run combinations so that all windows from a given run were assigned to the same fold. In both evaluation schemes, hyperparameter tuning was carried out within the respective cross-validation procedure on the training–validation set only; the selected configuration was then refitted on the full training–validation data before testing.

Three tree-based regression algorithms were used for model training and evaluation: Random Forest (RF), XGBoost (XGB), and CatBoost (CB).

Barry *et al.* [8] implemented the same models on this dataset using an ad-hoc approach to predict performance error; the present study extends this work by systematically examining real-time forecasting performance across feature configurations (lag vs. no-lag) and generalization strategies. For each algorithm, a discrete hyperparameter grid was specified to reflect key design choices (e.g., number of trees or boosting iterations, maximum tree depth, and column subsampling rate), and hyperparameter optimization was performed using exhaustive grid search with cross-validation. The negative root mean squared error, averaged across output variables, served as the selection criterion for the best estimator. The explored hyperparameters and their candidate values are summarized in Table 1.

Table 1: Candidate model hyperparameters

Model	Hyperparameter	Values
Random Forest	Number of Estimators	[50, 100, 150]
	Minimum Samples per Split	[2, 3, 4]
	Minimum Samples per Leaf	[2, 3, 4]
	Bootstrapping	[True, False]
XGBoost	Number of Estimators	[10, 100]
	Maximum Depth	[2, 3, 4, 5]
	Columns Sampled by Tree	[0.3, 0.4, 0.5]
CatBoost	Maximum Depth	[2, 3]
	Learning Rate	[0.01, 0.02, 0.03, 0.05]
	L2 Regularization Coefficient	[6, 10]

4.4 *Evaluation Metrics*

Model performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination R^2. RMSE quantifies prediction error in the same units as the target variables and places greater weight on larger errors [8], whereas MAE provides a complementary, more robust measure of typical error magnitude that is less sensitive to outliers. R^2 captures the proportion of variance in each target explained by the model and serves as the primary indicator of overall model fit. For each cross-validation setting, the mean and standard deviation of these metrics across folds are reported, along with an overall mean aggregated across targets.

Furthermore, to assess whether the models preserved the relationships between specific error components and overall performance, Pearson correlations between component errors (glide slope, localizer, airspeed) and total error were examined for both true values and model predictions. These correlations were computed across all test samples within each evaluation setting and are reported alongside the primary error metrics.

4.5 *Implementation*

All preprocessing, modeling, and analysis code was implemented in Python using NeuroKit2, pandas, NumPy, and scikit-learn for data handling and the baseline model, XGBoost and Catboost for gradient boosting models, and Matplotlab/seaborn for visualization. Model explainability was examined using SHapley Additive exPlanations (SHAP) to inspect feature contributions for selected outputs. For each final model, SHAP values were computed on a subsample of 1,000 time points, and summary plots as well as local waterfall plots were generated to identify which features most strongly influenced the predicted performance errors.

5 RESULTS

This section reports the performance of three tree-based models for task performance prediction. First, overall subject-level and temporal generalization performance relative to the Random Forest baseline is summarized; next, error correlations and feature importance are examined.

5.1 *Subject-level Generalization*

In this section, how well the three models generalize to unseen pilots in the subject-level evaluation is presented, first in terms of average performance across subjects and outputs, then in terms of individual error types and per-subject variability.

Table 2 presents the mean MAE, RMSE, and R^2 across outputs and subjects for the Random Forest baseline, XGBoost, and CatBoost, with and without lagged features. In the without-lagged-features condition, all models generate negative overall R^2 values, indicating performance below a simple mean predictor, with Random Forest showing the worst fit and CatBoost the best of the tree. When lagged features are included, overall R^2 increases to approximately 0.96 for Random Forest and XGBoost and to 0.97 for CatBoost, accompanied by marked reductions in MAE and RMSE.

Per-output performance for the best subject-level model (CatBoost with lagged features) is reported in Table 3. For glide slope and localizer errors, R^2 values of approximately 0.98 indicate that the model captures most of the between-subject variance in these errors. Airspeed error is slightly more difficult to predict, with R^2 values around 0.96, whereas total error shows somewhat lower but still high explained variance ($R^2 \approx 0.97$). The corresponding Figure 5 in Appendix A shows that predicted values cluster tightly around the identity line for all outputs, confirming that both the direction and the magnitude of subject-level deviations are reproduced well.

Table 2: Overall subject-level generalization performance of the models, with and without lagged features.

Model	Mean MAE	Mean RMSE	Mean R^2
Random Forest (no-lag)	0.71	0.97	-1.21
Random Forest (lag)	0.07	0.12	0.96
XGBoost (no-lag)	0.69	0.91	-0.83
XGBoost (lag)	0.07	0.12	0.96
CatBoost (no-lag)	0.67	0.89	-0.67
CatBoost (lag)	0.05	0.10	0.97

Figure 1 compares the RMSE of total error per subject for the Random Forest and CatBoost models with lagged features. For all subjects, CatBoost generally achieves slightly lower RMSE than Random Forest. The most noticeable gains occur for subjects exhibiting relatively high baseline errors. For most pilots, changing the model architecture offers only limited additional benefit beyond the inclusion of lagged features. For glide slope and localizer errors, the mean RMSEs in the subject-level setting are low (approximately 0.07–0.08), but the corresponding standard deviations are relatively large (approximately 0.05–0.09). This pattern reflects substantial variability across pilots: some subjects show very small errors, whereas others retain noticeably higher prediction errors, indicating that generalization quality is not uniform across individuals.

Table 3: Per-output performance for CatBoost with lagged features (subject level generalization).

Output	MAE	RMSE	R^2
Glide slope	0.03±0.01	0.07±0.05	0.98±0.03
Localizer	0.03±0.02	0.08±0.09	0.98±0.02
Airspeed	0.07±0.01	0.12±0.02	0.96±0.04
Total	0.08±0.03	0.14±0.04	0.97±0.03

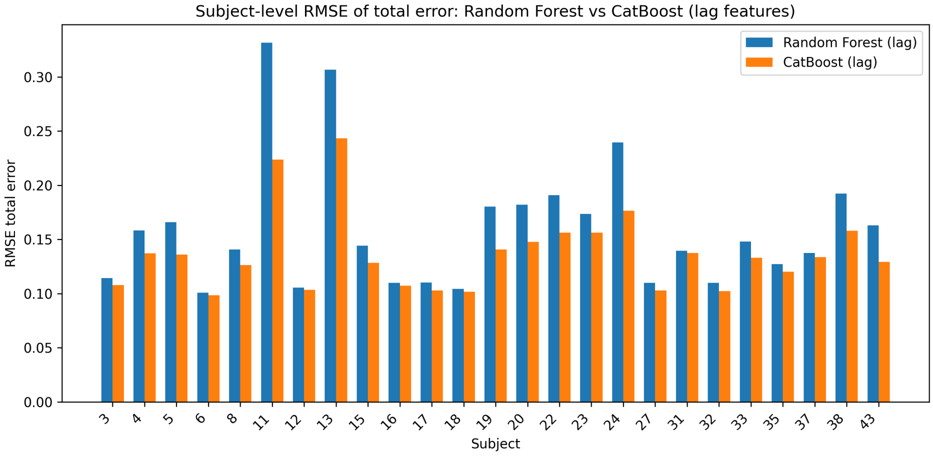

Figure 1: Comparison of total-error RMSE per subject for Random Forest and CatBoost models with lagged features

5.2 *Temporal Generalization*

Table 4 presents temporal generalization performance, averaged across outputs for the three models, with and without lagged features. In the without-lagged-features condition, all models achieve modest predictive accuracy, with overall R^2 values around 0.20, and RMSE close to 1.17. Including lagged features has a substantial improvement across models: overall R^2 increases to approximately 0.97 for Random Forest and to about 0.98 for XGBoost and CatBoost, while MAE and RMSE decrease by about a factor of ten. Among the three models, CatBoost with lagged features attains the lowest overall errors (MAE $\approx$ 0.07, RMSE $\approx$ 0.16).

Table 4: Overall temporal generalization performance of the models, with and without lag features

Model	Mean MAE	Mean RMSE	Mean R^2
Random Forest (no-lag)	0.68	1.13	0.20
Random Forest (lag)	0.09	0.19	0.97
XGBoost (no-lag)	0.75	1.17	0.18
XGBoost (lag)	0.09	0.18	0.98
CatBoost (no-lag)	0.73	1.15	0.20
CatBoost (lag)	0.07	0.16	0.98

Temporal performance per output for the three models in the lag condition is detailed in Table 5 . For glide slope and airspeed errors, all models achieve high explained variance, with R^2 above 0.97 and RMSE below 0.20, indicating accurate prediction of future deviations within runs. Localizer error is slightly more difficult to predict, with R^2 of approximately 0.96, although absolute errors remain small. CatBoost performs best overall, showing higher R^2 and lower MAE and RMSE than the other two models.

Table 5: Temporal performance per output of three models in the lag condition

Model / Metrics	Glide slope	Localizer	Airspeed	Total Error
RF MAE	0.09	0.07	0.09	0.12
RF RMSE	0.20	0.21	0.15	0.21
RF R^2	0.98	0.95	0.98	0.99
XGB MAE	0.08	0.06	0.07	0.13
XGB RMSE	0.18	0.19	0.13	0.23
XGB R^2	0.98	0.96	0.98	0.99
CB MAE	0.06	0.05	0.06	0.10
CB RMSE	0.15	0.18	0.11	0.19
CB R^2	0.99	0.96	0.99	0.99

The comparison between without-lagged-features and with-lagged-features conditions in Figure 2 highlights the role of short-term history in temporal prediction. For example, the RMSE of CatBoost for total error decreases from approximately 1.59 to 0.19 when lagged features are added, and the corresponding R^2 increases from 0.34 to 0.99. Similar improvements are observed for Random Forest and XGBoost. These results suggest that including preceding error is essential for accurate forecasting of near-future task performance.

Comparing Tables 2 and 4 shows that generalization to unseen pilots is slightly more difficult than temporal prediction within runs. In the with-lagged-features condition, all models achieve higher R^2 for temporal prediction than for subject-level prediction. This pattern suggests that individual pilots exhibit consistent short-term dynamics that can be exploited to anticipate upcoming deviations, whereas cross-subject variability remains more challenging to model.

5.3 *Error Correlations and Feature Importance*

Table 6 reports Pearson correlations between true component errors and true total error. The true data show only weak associations between individual component errors and total error, indicating that overall error combines several largely distinct parts rather than being dominated by a single component. Correlations between predicted individual errors and true total error are presented in Table 10 in Appendix A. In the subject-level

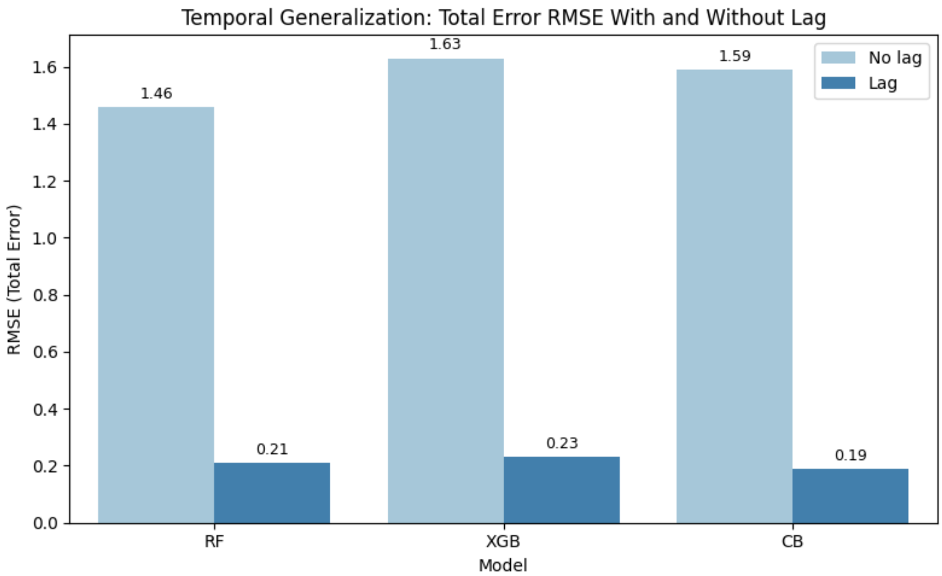

Figure 2: Comparison of total-error RMSE in temporal generalization for three models with and without lagged features

and temporal without-lagged-features conditions, these correlations are small and do not match the corresponding correlations observed in the true data, and predicted total error correlates only moderately with the true values, with Pearson correlation coefficient ranging from approximately 0.22 to 0.70 depending on model and generalization setting.

Table 6: Pearson correlations between true component errors and true total error

	Pearson Correlation	
Component Error	Subject-level	Temporal
Glide Slope	0.09	0.10
Localizer	-0.20	-0.10
Airspeed	0.22	0.20

In contrast, in the with-lagged-features conditions, all three models match the observed correlation pattern more closely. For both subject-level and temporal generalization, predicted component errors exhibit correlations with total error that closely resemble the true correlations. The correlation between predicted and true total error reaches 0.99 across models. These results indicate that the models achieve accurate predictions of total error and, at the same time, reproduce the observed relationships between the individual error components and overall approach performance.

Figure 3 and 4 illustrates these relationships for the CatBoost model under the with-lagged-features condition, showing that the regression lines for the predicted component errors closely follow those of the true errors across the range of total error.

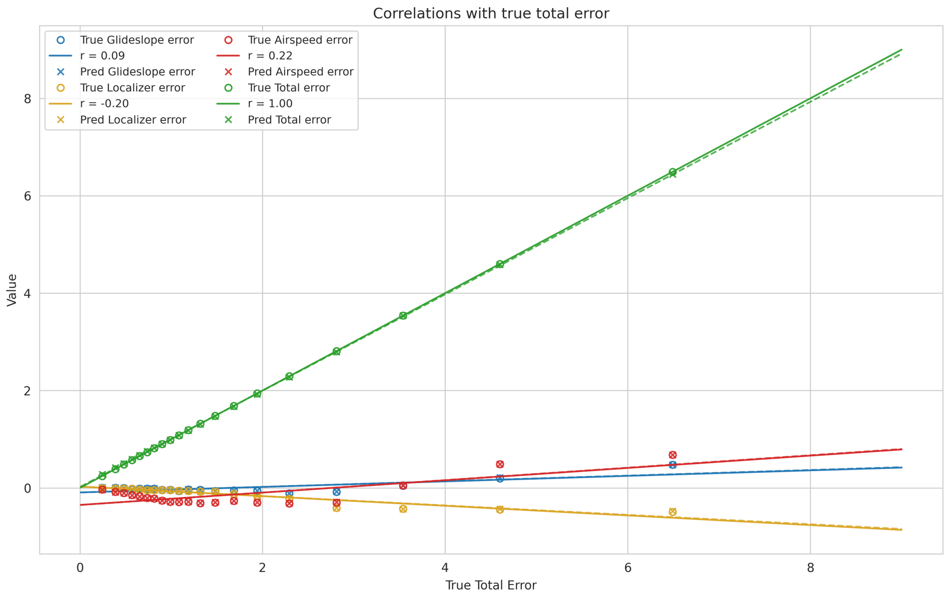

Figure 3: Correlations between component errors and the true total error for CatBoost (subject-level generalization with lagged features)

Feature importance analyses for the models in the without-lagged-features condition, as shown in Table 11 and 12 in Appendix A, indicate that elevation, aileron, and physiological variables related to heart rate and respiration are the most influential predictors of the performance variables. For example, Random Forest and XGBoost assign the highest importance to mean elevation trim and mean aileron trim, followed by heart-rate range and breathing-related features.

When lagged features are included, the importance pattern changes substantially, as presented in Table 7 and 8. In all three models, lagged performance variables become the primary predictors. The first lag of total error receives the highest importance, followed by the first lags of glide slope, airspeed, and localizer errors. The second lags and sensor features have much smaller importance scores. This pattern indicates that recent performance history is the strongest predictor of future errors, while elevation trim, and physiological variables have additional but relatively modest effects.

SHAP analysis for the CatBoost model in the temporal generalization setting with lagged features confirms the feature-importance findings.

Table 7: Feature importance for subject-level generalization (lag condition). Only the most influential lag features are shown.

Feature	Random Forest	XGBoost	CatBoost
Total Error (lag1)	0.521	0.019	80.338
Total Error (lag2)	0.001	0.173	19.139
Airspeed Error (lag2)	0.001	0.054	0.171
Airspeed Error (lag1)	0.142	0.178	0.159
Glide Slope Error (lag1)	0.199	0.241	0.051
Glide Slope Error (lag2)	0.002	0.056	0.051
Localizer Error (lag1)	0.126	0.087	0.026
Localizer Error (lag2)	0.005	0.162	0.012

The SHAP summary plot (Figure 6 in Appendix A) shows that lagged total error from the preceding one second has the largest impact on the predicted total error, followed by lagged total error from the previous two seconds and the lagged airspeed, localizer, and glide slope errors, which is consistent with the dominance of lagged performance variables in the importance rankings. High values of the lagged total error are associated with positive SHAP values, indicating increased predicted total error, whereas low values of this lagged error yield negative SHAP values and thus reduce the prediction. Physiological features such as HR range and mean BR exhibit much smaller SHAP magnitudes, indicating that these variables play only a minor role compared to lagged performance errors in the CatBoost model.

Table 8: Feature importance for temporal generalization (lag condition). Only the most influential lag features are shown.

Feature	Random Forest	XGBoost	CatBoost
Total Error (lag1)	0.524	0.022	89.834
Total Error (lag2)	0.001	0.153	9.647
Airspeed Error (lag2)	0.001	0.038	0.191
Airspeed Error (lag1)	0.152	0.185	0.174
Localizer Error (lag1)	0.126	0.099	0.031
Glide Slope Error (lag2)	0.019	0.067	0.024
Glide Slope Error (lag1)	0.168	0.233	0.022
Localizer Error (lag2)	0.005	0.167	0.012

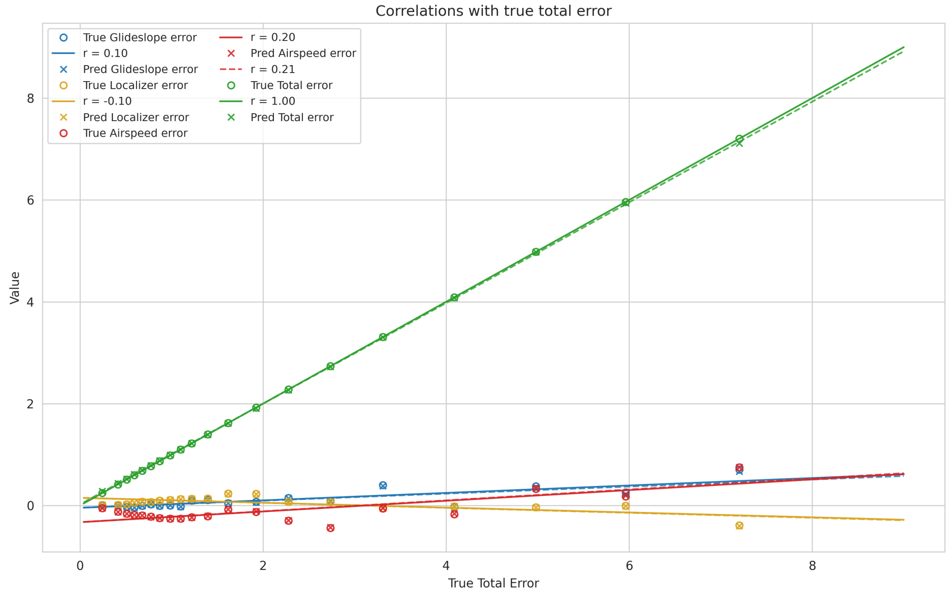

Figure 4: Correlations between component errors and the true total error for CatBoost (temporal generalization with lagged features)

6 DISCUSSION

This section presents an evaluation of the results obtained from the three tree-based regression models for forecasting performance errors, and discusses the limitations of the study as well as directions for future research.

6.1 *Main Findings*

This thesis examined how accurately pilots' performance during virtual reality landing tasks can be predicted in near real time using tree-based regression models trained on multimodal physiological and simulator-derived features. In both subject-level and temporal generalization settings, models without lagged flight-performance features achieved limited predictive accuracy: in the subject-level setting, all models yielded negative R^2 values, values, indicating performance below a mean-baseline predictor, and in the temporal setting, overall R^2 reached only about 0.20. In contrast, adding one- and two-step lagged error features led to high explained variance for all three models, with R^2 increasing to approximately 0.97 in the subject-level setting and to 0.98 in the temporal setting, and MAE and RMSE decreasing to roughly one-tenth of their values in the corresponding without-lagged-features condition.

Differences between the three models were relatively small compared with the gains obtained from including lagged features. CatBoost consistently achieved the lowest MAE and RMSE and the highest R^2 values, but Random Forest and XGBoost showed only negligible differences on the same evaluation metrics, suggesting that the inclusion of recent performance history is more critical than the specific model architecture. Comparing results from the two generalization methods further revealed that temporal prediction within runs is slightly easier than generalization to unseen pilots: all models achieved higher R^2 in the temporal setting than in the subject-level setting, indicating that individual pilots exhibit consistent short-term dynamics that are easier to predict than cross-subject variability.

Analyses of error correlations showed that, in the true data, component errors were weakly associated with total error, indicating that overall landing performance reflects a combination of distinct error sources rather than being dominated by a single component. When lagged performance errors were included, all three models closely replicate this correlation pattern, and high correlation between predicted and true total error, indicating that the models not only achieved accurate point predictions but also reserved the relationships among error components. Feature-importance and SHAP analyses further indicated that lagged performance variables dominated the predictions in both generalization settings, whereas physiological features had much smaller effects.

6.2 *Relation to Prior Work*

Prior studies in simulated and VR flight paradigms have shown that ECG-derived heart rate and respiration measures are sensitive to mental workload, stress, and overall performance [19, 20, 21]. Consistent with these findings, the present models assigned non-zero importance to certain HR- and respiration-based features; however, their contributions were clearly outweighed by lagged performance variables and simulator-derived features in the temporal generalization setting. This pattern suggests that recent error and control history already capture much of the information needed for short-horizon prediction, leaving comparatively little unique variance to be explained by physiology.

The present findings also extend earlier studies conducted on the same dataset that focused on flight-difficulty classification or offline landing-error regression. Tree-based and deep learning models have been shown to classify discrete difficulty levels from subsets of physiological and eye-tracking signals [7, 12], and Barry *et al.* [8] demonstrated that tree-based models trained on summary features can predict aggregated landing-error metrics. The current results go beyond this work by showing that tree-based multi-output regressors, equipped with simple lagged performance features, can achieve high explained variance for continuous landing-error trajectories at a one-second prediction horizon in both subject-level and temporal generalization settings.

6.3 *Methodological Considerations and Limitations*

The generalisability of these findings is constrained by several methodological choices. First, all analyses were conducted on a relatively small sample of participants performing a single VR ILS landing task, which may limit the extent to which the models transfer to other training scenarios or populations. Second, the models focused on very short time-window prediction using one- and two-step lagged error and feature values; longer-horizon forecasting may reveal a stronger role for physiological dynamics, as McGuire *et al.* [25] showed that physiological signals can estimate errors up to 10 seconds before they occur in real-world flight operations. Third, the feature engineering pipeline used manually designed summary statistics and lagged features instead of end-to-end temporal models that learn directly from raw time-series data. Finally, although the study examined both cross-subject and temporal generalization, the models were not tested in scenarios where they provide feedback to human instructors during training.

6.4 *Implications and Future Directions*

Despite these limitations, the strong predictive power of lagged error terms suggest that relatively simple models, deployed with simulator metrics, could provide reliable short-term forecasts of performance pattern and thus allow automated feedback to instructors. The comparatively smaller contributions of physiological features indicate that physiology can complement performance history to monitor workload and stress. More broadly, the modeling framework illustrates how multimodal data form VR training devices can be integrated into interpretable, real-time error-prediction pipelines, providing a step towards intelligent instructor systems that support scalable, objective assessment.

Future research could extend this work in multiple directions. First, evaluating models for longer-horizon performance prediction would provide insight into how far ahead performance errors can be forecast, and whether physiological dynamics contribute more strongly when predicting several seconds into the future. Second, applying the modeling framework to other VR flight tasks or datasets, including scenarios with different levels of difficulty, environmental conditions, or trainee expertise, would help assess robustness and external validity. Finally, integration of such models into live VR training sessions could help study their impact on instructor workload, trainee learning curves, and relevant safety behavior while providing empirical evidence on whether physiology-informed performance prediction meaningfully improves training efficiency and operational safety outcomes.

7 CONCLUSION

This thesis showed that tree-based regression models can predict second-by-second glide slope, localizer, airspeed, and total error in a VR landing task with high accuracy when one- and two-second lagged performance errors are provided in addition to ECG- and respiration-derived features and simulator variables. In both subject-level and temporal generalization settings, models without lagged errors did not display reasonable predictive power, whereas adding short-term history features greatly increased explained variance and sharply reduced prediction errors. The analyses further revealed that recent performance history is the dominant predictor of near-future errors, with physiological features providing useful but comparatively limited additional information. Together, these findings indicate that tree-based regression models with lagged inputs are capable of highly accurate, near real-time monitoring of pilot performance during training, supporting the development of adaptive VR training systems that can ease instructor workload and enable more scalable pilot training. At a broader level, the results suggest that, for very short-horizon forecasting of complex skills, recent performance history may carry most of the predictive power, while physiology may become more important when examining longer-horizon or more state-oriented performance prediction.

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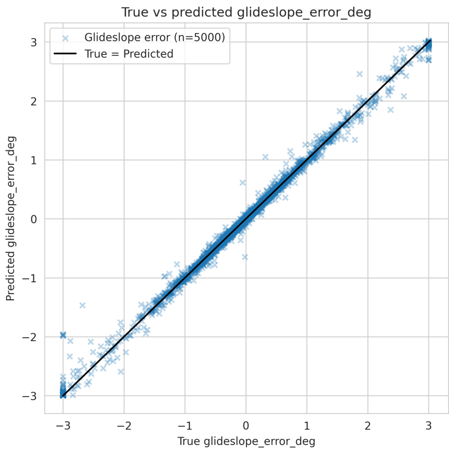
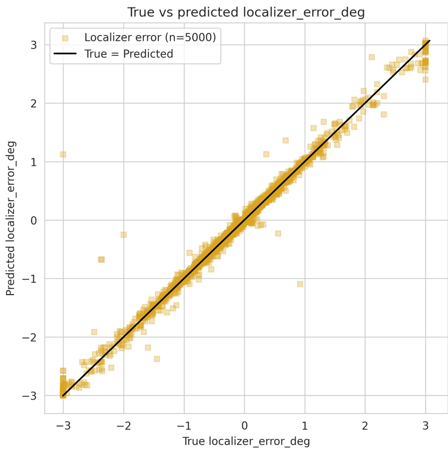
APPENDIX A

Table 9: Multimodal physiological data

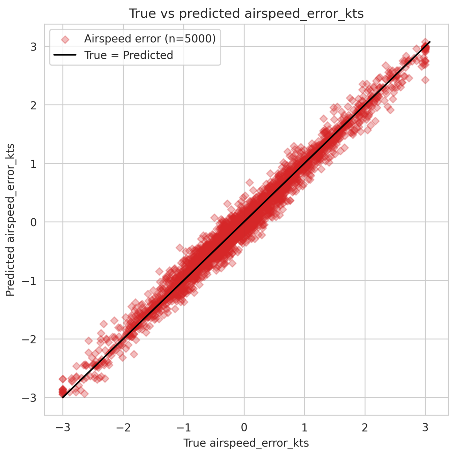
Variable Name	Unit	Description
EMG Wrist Flexor	mV	EMG of wrist flexor muscles
EMG Wrist Extensor	mV	EMG of wrist extensor muscles
PPG Finger	mV	Optical signal at tip of the middle finger
EDA Hand L	KOhms	EDA measured on the left hand
ECG Projection LL-RA	mV	ECG Signal: LL-RA projection
ECG Projection LA-RA	mV	ECG Signal: LA-RA projection
ECG Projection VX-RL	mV	ECG Signal: VX-RL projection
Respiration Trace	mV	Respiration Trace
Accelerometry Forearm x/y/z	m/s^2	Accelerometry of the right forearm
Accelerometry Torso x/y/z	m/s^2	Accelerometry of the torso
Gaze Origin L/R x/y/z	mm	Origin point of the gaze ray from each eye
Gaze Direction L/R x/y/z	mm	The normalized gaze direction of each eye
Eye Openness L/R	-	A value representing how open each eye is
Pupil Position L/R x/y/z	-	Normalized position of each pupil in sensor area
Pupil Diameter L/R	mm	Diameter of each pupil



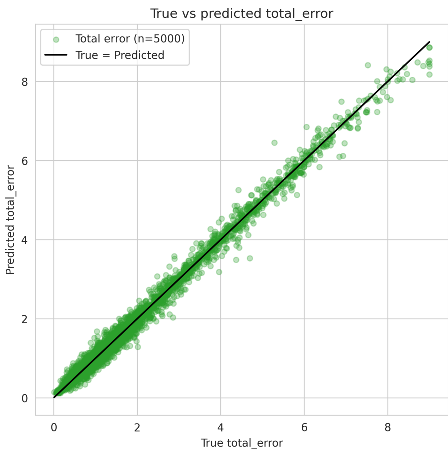
(a) Glide slope



(b) Localizer



(c) Airspeed



(d) Total error

Figure 5: True–versus–predicted errors for the best subject-level model.

Table 10: Pearson correlations between predicted individual errors and true total error

		Pearson Correlation	
Individual Error	Model	Subject-level	Temporal
Glide Slope	RF (no-lag)	-0.07	0.08
	RF (lag)	0.09	0.09
	XGB (no-lag)	-0.07	0.16
	XGB (lag)	0.09	0.10
	CB (no-lag)	-0.10	0.16
	CB (lag)	0.09	0.10
Localizer	RF (no-lag)	0.04	0.01
	RF (lag)	-0.20	-0.11
	XGB (no-lag)	0.04	-0.14
	XGB (lag)	-0.20	-0.11
	CB (no-lag)	-0.01	-0.08
	CB (lag)	-0.20	-0.10
Airspeed	RF (no-lag)	0.11	0.17
	RF (lag)	0.23	0.22
	XGB (no-lag)	0.04	0.17
	XGB (lag)	0.22	0.21
	CB (no-lag)	0.05	0.18
	CB (lag)	0.22	0.21
Total Error	RF (no-lag)	0.28	0.70
	RF (lag)	0.99	0.99
	XGB (no-lag)	0.24	0.61
	XGB (lag)	0.99	0.99
	CB (no-lag)	0.22	0.64
	CB (lag)	0.99	0.99

Table 11: Feature importance for subject-level generalisation without lag features. Only the most influential non-lag features are shown; importance scores are the model-specific values reported by each algorithm.

Feature	Random Forest	XGBoost	CatBoost
Mean Elevation	0.247	0.222	33.716
HR Range	0.094	0.062	18.277
Mean Aileron	0.127	0.138	12.434
Difficulty Level	0.053	0.134	10.610
Mean HR	0.066	0.036	4.601
Mean BR	0.075	0.050	4.426
SD BR	0.054	0.046	3.813
Max HR	0.059	0.117	5.350
Min HR	0.078	0.052	1.206
SD HR	0.035	0.066	0.418
Mean RESP Amp	0.071	0.052	5.150
SD RESP Amp	0.042	0.026	0.000

Table 12: Feature importance for temporal generalization without lag features. Only the most influential non-lag features are shown; importance scores are the model-specific values reported by each algorithm.

Feature	Random Forest	XGBoost	CatBoost
Mean Elevation	0.248	0.154	30.065
HR Range	0.089	0.066	11.620
Mean Aileron	0.131	0.112	14.929
Difficulty Level	0.056	0.136	9.593
Mean HR	0.062	0.059	7.272
Mean BR	0.075	0.050	6.898
SD BR	0.055	0.048	4.088
Max HR	0.061	0.010	5.102
Min HR	0.072	0.051	3.417
SD HR	0.036	0.146	0.614
Mean RESP Amp	0.074	0.046	6.259
SD RESP Amp	0.041	0.033	0.143

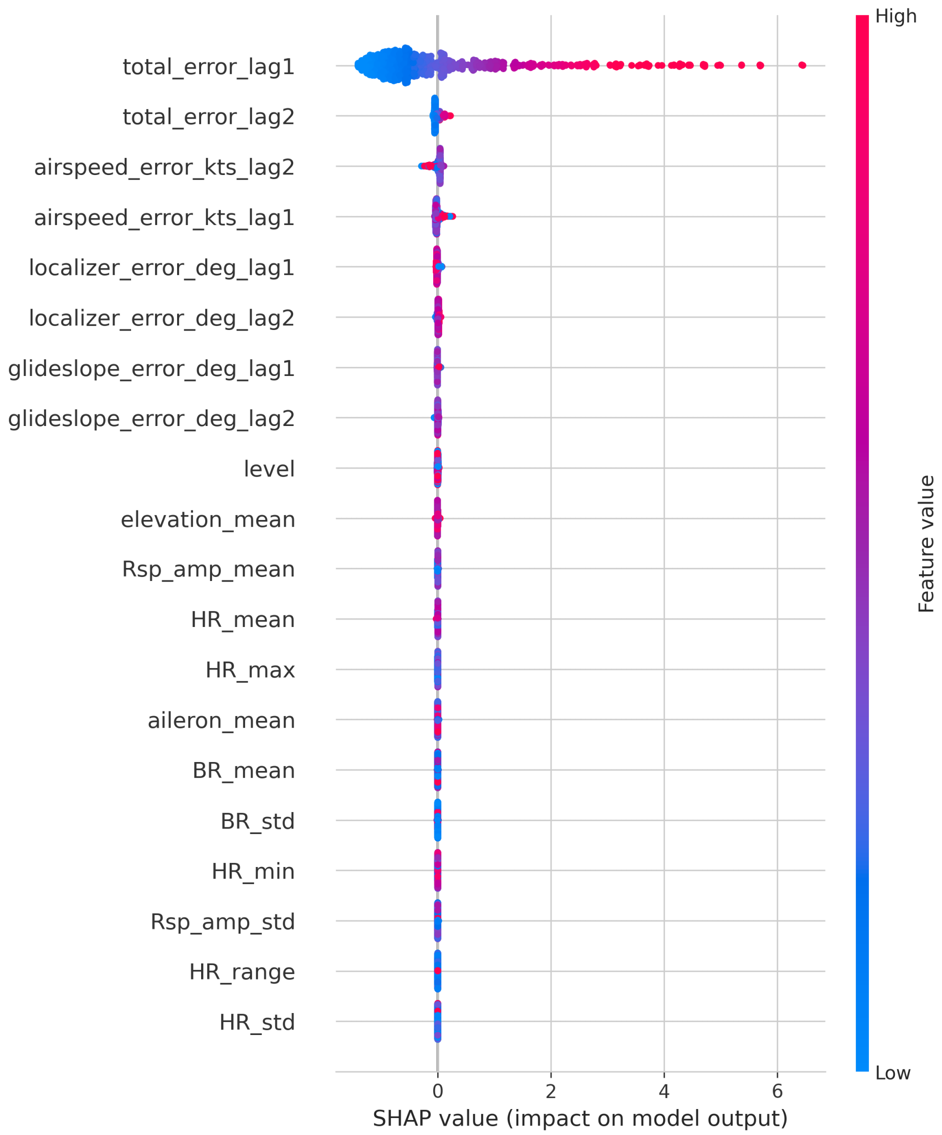


Figure 6: SHAP summary plot for CatBoost (temporal generalization with lag features)